THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE CHAPTER TEST SET B

Chapter 10: **Light: Shadows and Reflections** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which is a natural source of light? 1

(a) Moon **(b)** mirror **(c)** Sun **(d)** book

2. Frosted glass is: 1

(a) luminous **(b)** transparent **(c)** opaque **(d)** translucent

3. A shadow is darker for: 1

(a) opaque objects **(b)** glass

(c) air **(d)** transparent objects

4. The image in a plane mirror is: 1

(a) real and erect **(b)** virtual and inverted

(c) virtual and erect **(d)** real and inverted

5. The word AMBULANCE is written laterally inverted so it: 1

(a) reads correct in a rear-view mirror **(b)** looks fancy

(c) is invisible **(d)** is shorter

6. A periscope uses: 1

(a) a prism only **(b)** two lenses

(c) two plane mirrors **(d)** one concave mirror

7. The Sun's light reaches Earth through: 1

(a) vacuum (space) **(b)** glass **(c)** air only **(d)** water

8. The angle of incidence equals the angle of: 1

(a) dispersion **(b)** absorption **(c)** reflection **(d)** refraction

9. A plane mirror is a: 1

(a) rough wall **(b)** transparent sheet

(c) flat reflecting surface **(d)** curved surface

10. Solar eclipse shadows are formed because the Moon is: 1

(a) luminous **(b)** translucent **(c)** opaque **(d)** transparent

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 15–25 words each.

11. State the law of reflection. 2

12. What is lateral inversion? 2

13. Why is AMBULANCE written laterally inverted on vehicles? 2

14. State two characteristics of the image in a plane mirror. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how a periscope works. 3

16. Explain how a kaleidoscope works. 3

17. Why do we see a clear image in a mirror but not on a wall? 3

OR

Explain why the image in a pinhole camera becomes blurred when the hole is enlarged. 3

18. Differentiate between a shadow and an image in a pinhole camera. 3

OR

An object is 30 cm in front of a plane mirror. Find the distance between the object and its image. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

19. (i) Describe the construction of a pinhole camera. 5
(ii) State the nature of its image.
(iii) Effect of hole size.

OR

(i) How does a periscope work? 5
(ii) Where is it used?
(iii) What are the mirror positions?

20. **Case study:** A child makes a kaleidoscope using three mirror strips. 5
(i) What does it produce?
(ii) Why?
(iii) What are mirrors' angles?
(iv) Give another mirror device.

OR

(i) State the law of reflection. 5
(ii) Describe the activity to verify.
(iii) What is regular reflection?

END OF QUESTION PAPER — Total: 40 Marks

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